

Probability Solutions

Math 252

4: Urn

```
(choose(3, 2) +      # pick 2 red balls
 choose(5, 2)) /     # or pick two blue balls
 choose(8, 2)        # pick 2 of the 8 balls
```

```
[1] 0.4642857
```

5: Dice

There are $6 \cdot 6 = 36$ possible equally likely rolls. Four of them sum to 5, so $4/36$.

6. Smaller Dice

$4/16$

7. 6 or Doubles

There are 6 ways to get doubles. There are 5 ways to get a 6 ($1 + 5, 2 + 4, 3 + 3, 4 + 2, 5 + 1$), but notice that $3 + 3$ is in both lists, so we have $6 + 5 - 1$ ways to get a sum of six or doubles.

Answer; $10/36$

8. Lotto

Number of lotto tickets: $10^3 = 1000$.

Probabiliyt of big prize: $1/1000$.

Probability of small prize: $6/1000 - 1/1000 = 5/1000$. (There are $3 \cdot 2 \cdot 1$ ways to order three different numbers, but one of those wins the big prize.)

If you pick 7-7-7, you can only win the big prize. (In the real lottery, your big prize is bigger to make up for not having a chance to win the smaller prize.)

9. Flush

```
choose(4, 1) *      # pick a suit
 choose(13, 5) /     # pick 5 cards of that suit
 choose(52, 5)      # pick 5 cards from teh 52-card deck
```

```
[1] 0.001980792
```

10. 4-of-a-kind

```
choose(13, 1) *      # pick a number to have 4 of
 choose(4, 4) *      # pick all four of that number
```

```
choose(12, 1) *      # pick a number to have 1 of
choose(4, 1) /      # pick 1 of those 4 cards
choose(52, 5)      # pick 5 cards from teh 52-card deck
```

```
[1] 0.000240096
```

11. Random Bit String

Only one string doesn't have any 0's. So using the complement rule, we have

```
1 - 1 / 2^8
```

```
[1] 0.9960938
```

12. Yahtzee.

All numbers match:

```
6 / 6^5
```

```
[1] 0.0007716049
```

At least 2 match = not all different, so

```
1 - (6 * 5 * 4 * 3 * 2 * 1) / 6^5
```

```
[1] 0.9074074
```

13. Loaded Dice

If six is twice as likely as the rest, solve $5p + 2p = 1$ to get the probability of rolling a 1. $p = 1/7$.

Probability of rolling a 6 is $2/7$.

Probability of rolling doubles: $5 \cdot \frac{1}{7} \cdot \frac{1}{7} + \frac{2}{7} \cdot \frac{2}{7} = 0.1837$

15. More Lottery

```
# a.
```

```
1 / choose(40, 6)
```

```
[1] 2.605266e-07
```

```
choose(6,6) * choose(34, 0) / choose(40, 6)
```

```
[1] 2.605266e-07
```

```
# b
```

```
choose(6, 5) * choose(34, 1) / choose(40, 6)
```

```
[1] 5.314742e-05
```

correct	numerator	probability
6	1	2.605e-07
5	204	5.315e-05
4	8415	0.002192
3	119680	0.03118
2	695640	0.1812
1	1669536	0.435
0	1344904	0.3504